

## AuthService – Complete A→Z Testing and Usage Guide

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This document provides a full testing workflow for your AuthService API including:

- Registration → Email Confirmation → Login
- 2FA Authentication Flow
- Refresh & Revoke Token
- Forgot/Reset/Change Password Flows
- Role-Based Authorization
- Address CRUD demonstrating CQRS Dual Database
- External OAuth Login Setup (Google/Microsoft)
- Health Check & Logging Verification

### 1. Running The Application

#### Option A - Docker Compose

```
docker compose up --build
```

Navigate to: <http://localhost:8080/swagger>

#### Option B - Local Execution

```
dotnet restore
```

```
dotnet build
```

```
dotnet run --project src/AuthService.Api
```

### 2. Initial Database Migration (If Required)

```
dotnet ef database update -c WriteDbContext -p src/AuthService.Infrastructure -s src/AuthService.Api
```

```
dotnet ef database update -c ReadDbContext -p src/AuthService.Infrastructure -s src/AuthService.Api
```

### 3. Testing the Authentication Flow (A→Z)

Step 1: Register

POST /api/auth/register

Body Example:

```
{  
  "email": "alice@example.com",  
  "password": "Passw0rd!",  
  "firstName": "Alice",  
  "lastName": "Doe"  
}
```

Step 2: Confirm Email

Retrieve confirmation link from logs and open in browser.

Step 3: Login

POST /api/auth/login

Returns:

- accessToken (JWT)
- refreshToken

Step 4: Access a Protected Endpoint

GET /api/users/me

Header:

Authorization: Bearer

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#### 4. Refresh Token Flow

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POST /api/auth/refresh

Body:

""

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#### 5. Two-Factor Authentication

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Enable 2FA:

POST /api/auth/2fa/enable

Disable 2FA:

POST /api/auth/2fa/disable

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#### 6. Password Flows

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Forgot Password:

POST /api/auth/password/forgot

Reset Password:

POST /api/auth/password/reset

Change Password (Authenticated):

POST /api/auth/password/change

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## 7. CQRS Dual Database Testing (Address CRUD)

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Create Address (SQL Server — WriteDbContext):

POST /api/users/addresses

List Addresses (PostgreSQL — ReadDbContext):

GET /api/users/addresses

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## 8. External OAuth Login Testing (Google / Microsoft)

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Update appsettings:

ExternalAuth.Google.ClientId & ClientSecret

ExternalAuth.Microsoft.ClientId & ClientSecret

Redirect URIs must match provider config:

http://localhost:8080/signin-google

http://localhost:8080/signin-microsoft

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## 9. Role-Based Authorization

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Admin User (Seeded):

Email: admin@demo.local

Password: Admin@12345

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## 10. Health Check and Logging

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GET /health

docker compose logs -f api

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End of Guide.